

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours

Maximum Marks:35

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- Q.1 Implement the C program to accept n integers to be sorted. Main function creates child process using fork system call. Parent process sorts the integers using bubble sort and waits for child process using wait system call. Child process sorts the integers using insertion sort [15 Marks]
- Q.2 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 Marks]
- Q.3. Oral/Viva [5 Marks]